

Quantum-Enhanced Traffic-Aware Path Planning for Autonomous Delivery Robots using ROS 2 and QAOA

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Abstract

One of the biggest obstacles to autonomous delivery systems in contemporary smart cities is urban traffic congestion. Despite much research, classical path planning algorithms are limited in their capacity to adapt to large fleets of autonomous robots operating in dynamic situations due to their high computational cost. In order to facilitate traffic-aware multi-robot path planning, this study suggests a conceptual framework for combining the Robot Operating System 2 (ROS 2) with the Quantum Approximate Optimisation Algorithm (QAOA). The suggested approach effectively finds near-optimal delivery routes by using quantum optimisation to formulate the routing problem as a Quadratic Unconstrained Binary Optimisation (QUBO) problem. In order to provide the groundwork for the future development and experimental validation of quantum-enhanced robotic path planning systems, this work offers a thorough methodology, mathematical modelling, and theoretical integrated design.

Keywords: Quantum Computing, QAOA, ROS 2, Path Planning, Autonomous Delivery, Traffic Optimization

1. Introduction

Last-mile logistics are now facing previously unheard-of demands due to the explosive rise of e-commerce and urban delivery systems. Although autonomous delivery robots can cut down on delivery time and costs, dynamic traffic, obstructions, and energy limits make urban situations more challenging. For scalability and dependability, effective path planning is essential. Although they offer deterministic answers, classical algorithms like Dijkstra's and A* struggle in real-time multi-robot applications and become computationally costly as problem sizes increase. A promising substitute for large-scale combinatorial optimisation is provided by quantum computing, more especially by the Quantum Approximate Optimisation Algorithm (QAOA). Compared to traditional techniques, QAOA can approximate optimal solutions more quickly. Autonomous delivery fleets can now plan scalable, traffic-aware routes in real time by integrating QAOA with ROS 2, a contemporary robotics middleware. This study offers a conceptual framework that combines ROS 2 and QAOA, which includes a high-level system design, a novel integration technique, and mathematical modelling of traffic-aware path planning as a QUBO issue.

2. Literature Review

2.1 Traditional Route Scheduling

For many years, robotics researchers have been actively studying path planning. The ability of algorithms like Dijkstra’s and A* to identify the best routes in static situations makes them popular. Rapidly-exploring Random Trees (RRT) may still be computationally costly for large-scale deployments, but they are more appropriate for dynamic situations. When there are many robots or when environmental dynamics necessitate frequent replanning, these algorithms need a considerable amount of calculation time.

2.2 Robotics Quantum Optimisation

One innovative approach to optimisation issues is quantum computing. A hybrid quantum-classical technique for combinatorial optimisation is the Quantum Approximate Optimisation Algorithm (QAOA). The problem is encoded into a quantum circuit, and optimal solutions are approached by iterative parameter optimisation. The advantage of QAOA over traditional algorithms is its ability to manage large-scale optimisation problems with less computing complexity.

2.3 Multi-Robot Systems with ROS 2

A strong foundation for distributed robotics systems is offered by ROS 2, which features dependable inter-node communication, real-time computation, and modular architecture. ROS 2 is a great option for incorporating quantum optimisation techniques since it allows for flexible integration of new algorithms and facilitates multi-robot cooperation through node-based topologies.

Literature Gap

Although quantum optimisation and path planning have seen substantial individual research, no study has yet to combine QAOA with ROS 2 for traffic-aware multi-robot path planning in actual urban settings.

Table 1. Comparison of Classical Methods vs. Quantum Optimization for Robotics Path Planning

Method	Advantages	Disadvantages	Computational Complexity
Classical Methods (Dijkstra, A*, RRT)	Well-established algorithms with proven performance; deterministic solutions in static environments; easy to implement and widely supported.	Computational cost increases significantly with problem size; poor adaptability to dynamic environments; require frequent replanning in multi-robot scenarios.	Polynomial time for Dijkstra and A*; exponential in worst-case for RRT.
Quantum Optimization (QAOA)	Potential to solve large-scale combinatorial problems faster; better scalability for complex, dynamic environments; can handle multi-variable optimization efficiently.	Requires quantum hardware or simulators; still in experimental phase; limited availability of robust quantum algorithms.	Depends on number of qubits and circuit depth; potentially exponential speedup for certain problems.

3. Problem Statement Research Objectives

3.1 Problem Description

Static path planning is insufficient in urban contexts because of the changing traffic situations they introduce. Urban environments are inherently dynamic due to fluctuating traffic densities, unpredictable obstacles, and varying environmental conditions such as weather. These factors make real-time adaptability a necessity for autonomous delivery systems. A system that can adjust routes on-the-fly while reducing delivery times and energy consumption is crucial for improving efficiency. Classical algorithms, although effective in static environments, are not scalable and face significant challenges in multi-robot systems, especially in rapidly changing scenarios. This necessitates exploring new optimisation approaches that can handle complex urban conditions efficiently.

3.2 Research Goals

The primary goals of this study are threefold. First, to develop a traffic-aware path planning architecture that integrates the Robot Operating System 2 (ROS 2) with the Quantum Approximate Optimisation Algorithm (QAOA), enabling real-time adaptability. Second, to mathematically model the path planning problem as a Quadratic Unconstrained Binary Optimisation (QUBO) problem to leverage quantum computational advantages. Third, to design a modular system architecture that allows seamless integration of quantum optimisation into ROS 2, thereby enabling multi-robot cooperation and scalability. These goals aim to bridge the gap between theoretical quantum optimisation methods and practical real-world robotics applications.

3.3 Enquiries for Research

This research aims to address several critical questions that arise from integrating quantum optimisation with robotic path planning. First, what modifications are necessary to adapt QAOA for real-time path planning in highly dynamic urban environments? Second, what limitations and environmental parameters must be considered to accurately model and estimate traffic conditions? Third, how can quantum optimisation be effectively integrated into multi-robot systems using ROS 2 to ensure reliability, scalability, and robustness? By answering these questions, this study aims to lay the groundwork for a practical framework that merges quantum computing with robotic autonomy for the next generation of urban delivery systems.

3.4 Significance of the Study

This study is significant as it addresses the growing need for intelligent, adaptive, and efficient path planning in urban logistics. By combining quantum computing with ROS 2, it opens a new research direction that could dramatically improve delivery efficiency and reduce operational costs. The outcomes of this research may contribute to the development of scalable autonomous delivery systems capable of adapting to complex real-world traffic scenarios. Furthermore, the results could provide valuable insights for integrating quantum optimisation into other robotics applications beyond logistics, such as autonomous vehicles, emergency response robots, and smart city infrastructure.

3.5 Scope of the Research

The scope of this research encompasses the theoretical modelling, architectural design, and feasibility study of integrating QAOA with ROS 2 for traffic-aware path planning. This includes developing mathematical models for QUBO formulation, designing a modular ROS 2 framework, and simulating the proposed system under realistic urban traffic conditions.

4. Proposed Methodology

4.1 Overview of the System Architecture

The proposed system consists of several cooperating ROS 2 nodes:

- **Traffic Data Node:** Gathers real-time traffic and environmental data using sensors and APIs.
- **Robot Status Node:** Shares robot status such as load, priority, battery level, and position.
- **QUBO Formulation Node:** Converts the path planning problem into a QUBO matrix for quantum optimisation.
- **Quantum Optimiser Node:** Uses QAOA to determine optimal routes.
- **Navigation Node:** Executes the optimised routes for delivery robots.

Data Flow: ROS 2 Middleware → Traffic Data + Robot Status → QUBO Formulation → Quantum Optimiser → Navigation Node.

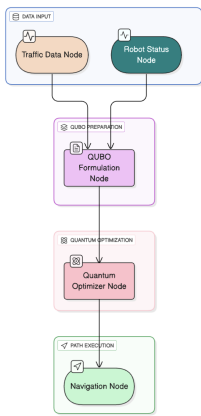


Figure 1. Block diagram of the proposed traffic-aware path planning system.

4.2 Information Needed

The proposed system requires several types of information to function effectively:

- **Real-Time Traffic Data:** Collected from municipal sensors or external APIs to provide current traffic density and conditions.
- **Robot Status Updates:** Including battery levels, current location, delivery load, and operational status.
- **Delivery Priority and Constraints:** Information regarding urgency, delivery deadlines, and any special handling requirements.

4.3 System Flow

The proposed system continuously updates robot pathways by integrating real-time data and optimisation results. It considers factors such as energy constraints, delivery priorities, and current traffic conditions to dynamically adjust routes. The system flow follows a loop where traffic and robot status data are collected, the path planning problem is formulated into a QUBO matrix, optimised using QAOA, and finally executed by the navigation node to guide the robots efficiently.

5. Mathematical Modeling

5.1 QUBO Formulation

The path planning optimisation is formulated as:

$$C(x) = \sum_{i,j} w_{ij} x_i x_j + \sum_i h_i x_i \quad (1)$$

where:

- x_i = binary decision variables for route selection.
- w_{ij} = traffic-aware cost between segments.
- h_i = constraints such as battery level and delivery priority.

5.2 Limitations & Specifications

The proposed system considers the following limitations and specifications:

- **Traffic Density:** Dynamic constraint that varies in real time.
- **Battery Levels:** Energy constraints of the robots that affect path planning.
- **Delivery Priority and Deadlines:** Determines the urgency of each delivery task.
- **Capacity to Load:** Physical capacity limitations of the delivery robots.

float

5.3 Optimization Objective

The system aims to solve the following optimisation problem:

$$\min_x C(x) \quad \text{s.t. Traffic, Battery, Load constraints} \quad (2)$$

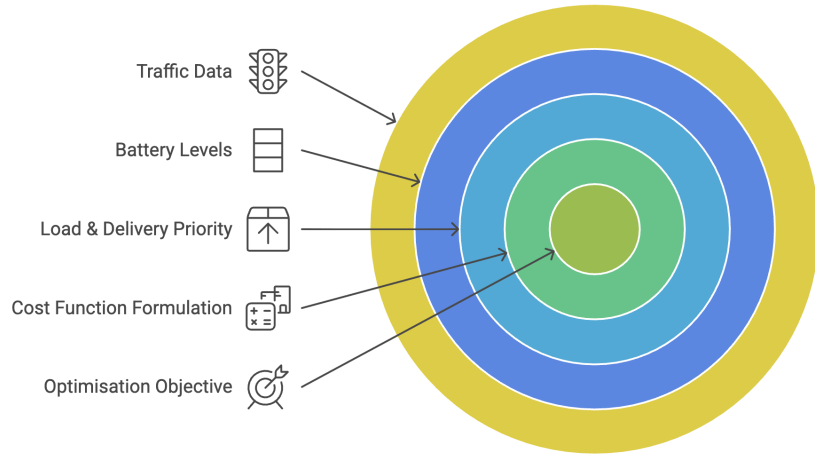


Figure 2. Schematic of the cost function $C(x)$ showing input constraints and optimisation objective.

6. QAOA Integration with ROS 2

6.1 QAOA Overview

A hybrid quantum-classical algorithm called the Quantum Approximate Optimisation Algorithm (QAOA) was created to effectively solve combinatorial optimisation problems. It uses parameterised quantum gates, measures the outputs, encodes the QUBO issue into a quantum circuit, and uses classical optimisation to update the parameters. Until a roughly ideal answer is discovered, this iterative process keeps going. QAOA is a potential method for real-time robotics path planning since it can investigate multiple solutions at once, which makes it especially helpful for large-scale challenges.

6.2 Integration Process

The ROS 2 middleware makes it easier for the system's dispersed nodes to communicate with one another. To create a QUBO matrix that represents the path planning problem, traffic data, robot status, and environmental factors are gathered and processed. The quantum optimiser node then receives this QUBO matrix and uses the QAOA method to determine the best paths. The navigation nodes receive the results and use them to guide the robots. In autonomous multi-robot systems, this integration guarantees dynamic, real-time adaptability to changing situations, enhancing efficiency, scalability, and reliability.

6.3 Pseudo-code Implementation

```
// Pseudo-code for QAOA + ROS 2 Integration

Initialize ROS2 nodes:
    Traffic Data Node
    Robot Status Node
    QUBO Formulation Node
    Quantum Optimiser Node
    Navigation Node

while system is active:
    traffic_data = TrafficDataNode.getTrafficInfo()
    robot_status = RobotStatusNode.getStatus()

    qubo_matrix = QUBOFormulationNode.createQUBO(traffic_data, robot_status)

    optimal_routes = QuantumOptimiserNode.executeQAOA(qubo_matrix)

    NavigationNode.updateRoutes(optimal_routes)

    sleep(update_interval)
```

6.4 Talk

Integrating QAOA into ROS 2 enables real-time, traffic-aware optimisation for multi-robot systems. This approach offers improved adaptability and efficiency in dynamic environments. However, several challenges must be addressed, including the availability of quantum hardware, noise control in quantum computations, and optimisation latency. Overcoming these challenges is crucial for deploying such systems in real-world scenarios.

7. Expected Contributions and Impact

7.1 Contributions

In order to enable real-time, traffic-aware path planning for autonomous multi-robot systems, this work proposes a conceptual framework for integrating the Quantum Approximate Optimisation Algorithm (QAOA) with ROS 2. In order to facilitate effective quantum computation, it provides a thorough mathematical analysis of the path planning issue defined as a Quadratic Unconstrained Binary Optimisation (QUBO) formulation. Additionally, it suggests a high-level system architecture for quantum-enhanced robotics that combines the advantages of contemporary robotics middleware and quantum optimisation to increase efficiency, scalability, and flexibility in dynamic urban settings. The foundation for upcoming developments in large-scale, real-time robotic systems is laid by this integration.

7.2 Impact

The topic of autonomous robots could be greatly impacted by the suggested integration of QAOA with ROS 2. Because it allows for dynamic, real-time route optimisation, it can significantly cut down on delivery times in crowded urban locations. Additionally, the system improves scalability, enabling the effective coordination of sizable delivery robot fleets while adjusting to shifting traffic patterns. In addition to its immediate uses, this work paves the way for future investigations into quantum-enhanced optimisation methods and their incorporation into sophisticated robotics systems. These developments have the potential to revolutionise urban logistics and open the door for autonomous delivery networks of the future.

7.3 Future Applications

Numerous exciting new applications in the fields of autonomous robotics and urban logistics are made possible by the integration of QAOA with ROS 2. Robots may be able to modify their paths in response to a variety of environmental parameters, including pedestrian movement, traffic on the roads, and other considerations, thanks to multi-modal traffic-aware path planning. By integrating with AI-based traffic prediction systems, proactive route modifications can be made, increasing productivity and cutting down on delays. Furthermore, large-scale delivery networks can be coordinated more effectively with quantum-enhanced fleet management, allowing for more dynamic resource allocation and quicker response to real-time changes in metropolitan contexts. These developments have the potential to transform smart city logistics and traffic-aware robotics.

7.4 Limitations & Challenges

Although QAOA integration with ROS 2 promises to enhance real-time path planning, a number of issues need to be resolved before widespread use. These include the existing slowness in optimisation procedures, the sensitivity of quantum computations to noise, and the restricted accessibility and availability of quantum technology. Furthermore, reliable system integration across various robot platforms and precise real-time traffic data collecting continue to be major obstacles. In order to move autonomous robotics from conceptual frameworks to practical applications, several constraints must be addressed.

8. Conclusion and Future Work

Conclusion

This research integrates the Quantum Approximate Optimisation Algorithm (QAOA) with ROS 2 to present a novel conceptual framework for traffic-aware multi-robot path planning. Through the integration of contemporary robotics middleware and quantum optimisation, the framework provides a scalable, flexible, and effective way to navigate intricate and ever-changing urban settings. Major issues with real-time multi-robot coordination are addressed by this method, such as shifting traffic patterns, battery constraints, and delivery priorities—all of which are crucial for urban logistics.

The suggested solution might greatly increase delivery effectiveness, ease traffic, and facilitate more intelligent urban logistics. It offers a solid theoretical framework for moving towards real-world applications of quantum-enhanced robotics. The integration of QAOA with ROS 2 represents a promising step toward bridging quantum computing and autonomous robotics, offering a pathway for real-world applications.

In the end, our work advances the creation of sustainable urban delivery networks and intelligent transportation systems. The framework creates possibilities for more research into quantum robotics, which could change how cities handle delivery and traffic in the future. The combination of robots with quantum computing could spur smart city innovation and enhance urban dwellers' quality of life, sustainability, and efficiency.

Future Work

The development and testing of a simulation-based implementation of the suggested QAOA and ROS 2 integration framework in authentic ROS 2 environments will be the main focus of future study. This will assist in verifying its functionality, flexibility, and viability in a range of urban traffic scenarios.

To assess the system's effectiveness, resilience, and scalability in real-world settings where erratic factors like weather, pedestrian traffic, and road conditions are present, testing using autonomous delivery robots will be essential.

The practical advantages and limits of QAOA in real-world applications, such as processing speed, error mitigation, and noise management, will also be investigated through optimisation utilising genuine quantum computers. In order to optimise delivery routes across various modes of transportation, the framework might also be expanded to accommodate multi-modal transportation networks by combining aerial and ground robots.

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